

HSE DATABASE MANAGEMENT SYSTEM

Transitioning to Digital Safety

Empowering the Oil and Gas Sector in the Albertine Graben

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EXECUTIVE SUMMARY

This project transitions HSE data management from fragmented, paper-based processes into a centralized **MySQL Relational Architecture**.

Strategic Focus:

- ✓ Tilenga & Kingfisher Project sites.
- ✓ NEMA & Petroleum Authority (PAU) Compliance.
- ✓ Mitigating high-risk operational hazards.



APPLICATION DOMAIN



Incident Management

Real-time logging of accidents, near-misses, and hazardous observations across remote rig sites.



Audit Tracking

Scheduling and recording environmental inspections at Buliisa and Hoima operational camps.



Compliance

Tracking BOSIET and First Aid certifications to ensure 100% field readiness of personnel.



User Groups: HSE Managers, Field Officers, HR Coordinators, and Auditors.

STRATEGIC BENEFITS ANALYSIS

Tangible Gains

- ✓ Fines reduction through PAU compliance.
- ✓ Lower insurance premiums via risk data.
- ✓ 80% reduction in monthly reporting time.

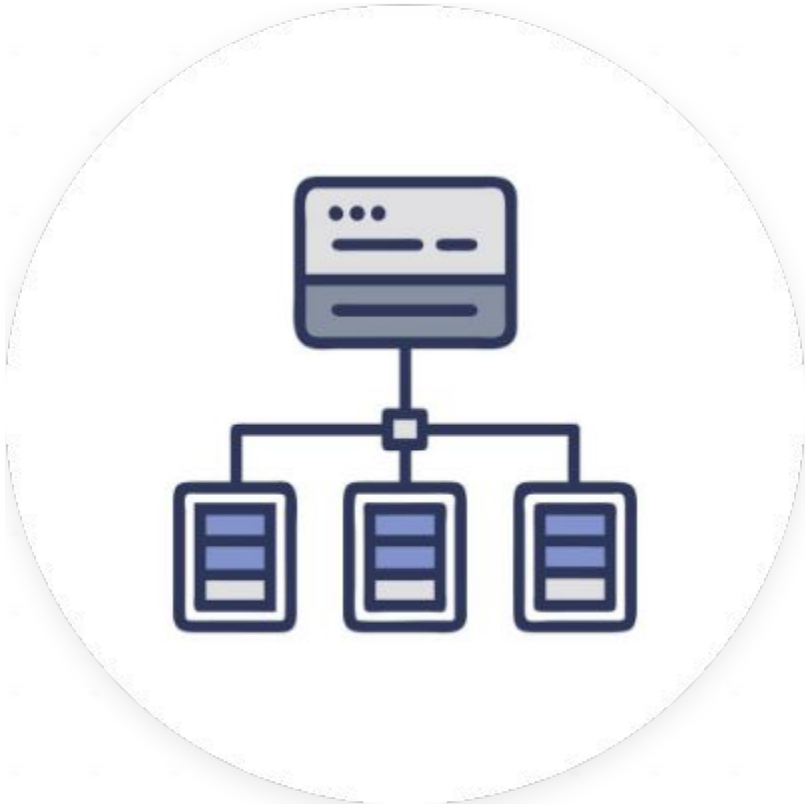
Intangible Value

- ✓ Enhanced corporate reputation in Uganda.
- ✓ Shift to a proactive "Safety First" culture.
- ✓ Increased employee trust and morale.

RISK ASSESSMENT & MITIGATION

- ✓ **Technical Connectivity:** Remote sites (Buliisa) face intermittent internet. Mitigation: Planned offline-sync capabilities.
- ✓ **Organizational Resistance:** Staff accustomed to legacy paper reporting. Mitigation: Localized Swahili/Luganda tooltips.
- ✓ **Data Integrity:** "Garbage in, Garbage out" principle. Mitigation: Strict input validation and mandatory fields.
- ✓ **Security Vulnerability:** Sensitive medical/incident data. Mitigation: Role-Based Access Control (RBAC).

LOGICAL SYSTEM DESIGN



Relational Entities

The system anchors all activities around the **Site** entity, branching into:

- ✓ **Site (1:M) Employees:** Operational tracking.
- ✓ **Site (1:M) Incidents:** Safety history.
- ✓ **Employee (M:N) Training:** Resolved via *Emp_Training* junction table.

PHYSICAL SCHEMA (MYSQL)

Entity	Primary Key	Attributes	Constraints
sites	site_id	site_name, location, risk_level	ENUM('Low', 'Med', 'High')
employees	emp_id	f_name, l_name, role, site_id	FK → sites(site_id)
incidents	incident_id	date, category, severity, site_id	FK → sites, FK → employees
trainings	course_id	course_name, validity_months	NOT NULL

Implementation

MySQL DDL & Data Population

UI & DIALOGUE STRUCTURE



Dashboard

KPI Widgets: Total Incidents, Pending Audits, and Expiring Certs.

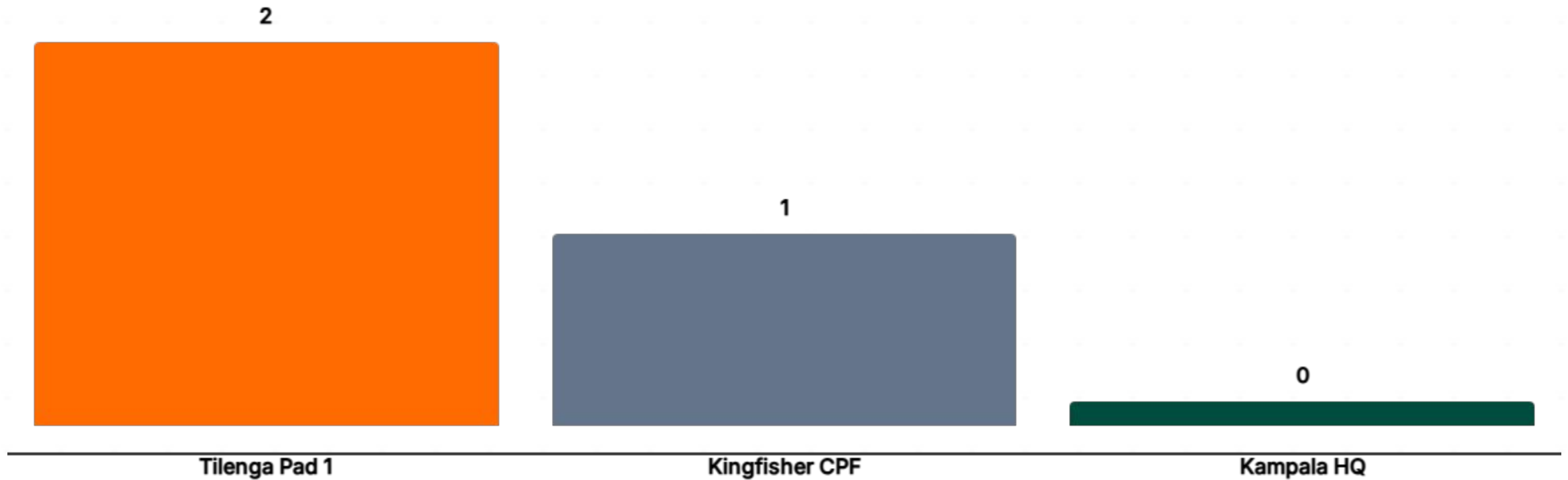
Incident Flow

Site Dropdown → Category (LTI, Spill) → Severity → Submit.

Reporting

Automated SQL aggregation for weekly PAU submissions.

INCIDENT FREQUENCY BY SITE



**Simulated Q4 2025 Data: Identifying high-risk drilling zones for immediate intervention.*



Conclusion & Future Scope

The HSE DBMS provides a functional repository for oil sector data. Future expansion includes **Offline-First** mobile sync and **AI Predictive Analytics** for hazard forecasting.

Thank You | Asimwe Gilbert | MUST 2026

END

Thank you for your time.

Bibliography

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